Prathamesh Palatshaha

25201310
Master of Technology
in Artificial Intelligence
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GitHub | LeetCode
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Education

Degree	Institute	University	CGPA/Percentage	Year
M.Tech AI	Dr B R Ambedkar National Institute of Technology,Jalandhar	NIT Jalandhar	8.52	2025-2027
B.E Computer Engineering	Vidyalankar Institute of Technology	University of Mumbai	9.39	2021-2025

Professional Experience

- **Innoshri Pvt Ltd** *April 2024 - July 2024*
  Machine Learning Intern Pune
  - Built automated fault detection system for Air Handling Units achieving 99.71% accuracy and 99.34% ROC-AUC by implementing Bagging Classifier with BorderlineSMOTE on LBNL's MZVAV-1 dataset from AFDD Evaluation Inventory using Python, Scikit-Learn, and imbalanced-learn
  - Optimized model performance achieving 99.94% precision and 99.92% recall (F1-score: 99.91%) through systematic evaluation of 8+ ensemble-sampling combinations, outperforming baseline Random Forest models by 16% in accuracy
  - Resolved 6:1 class imbalance affecting 14+ operational features by applying stratified K-fold splitting, robust scaling for 4 skewed columns (Supply Air Temp, OA Temp, RA Temp, Cooling Coil Signal), and min-max normalization using Pandas and NumPy
  - Engineered predictive maintenance framework processing minute-by-minute sensor data from multi-zone VAV systems (supply/return fans, cooling/heating coils, dampers) to enable early fault detection and minimize system downtime

Projects

- **ChatBOT - RAG Based QNA Bot**
  Personal Project **GitHub**
  - Built a PDF-based Question Answering system using RAG for querying document content
  - Enabled multi-modal extraction (text, tables, images) with vector-based semantic retrieval
  - Generated context-aware answers using an LLM (Gemini) grounded in PDF data
- **Banking Data Analysis**
  Data Analysis Project **GitHub**
  - Performed comprehensive analysis on banking data to extract key insights and trends
  - Cleaned, processed, and visualized datasets using Python (pandas, matplotlib)
  - Created interactive dashboards and statistical summaries to support business decision-making

Technical Skills

- **Programming Languages**: Proficient: Python, SQL. Familier: C, C++, Java, HTML, CSS
- **Tools and Frameworks**: Scikit-Learn, PyTorch, TensorFlow, FastAPI, Git, GitHub, PyCharm, VS Code, Jupyter Notebook/Lab, Google Colab, Linux, VirtualBox, HuggingFace, Langchain, Power BI
- **Core Subjects**: Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Generative AI, Prompt Engineering

Publications

- **Novel Approach for Predictive Maintenance of Air Handling Units Using Machine Learning** *April 2025*
  Journal of Neonatal Surgery **Link**
  - Developed an ML-based predictive maintenance system for Air Handling Units (AHUs) using LBNL's MZVAV-1 dataset, evaluating 10+ ensemble and Random Forest variants under multiple data-splitting strategies
  - Achieved **99.71% accuracy** using Bagging Classifier with BorderlineSMOTE, outperforming baseline Random Forest models by **16%** in accuracy and **10%** in F1-score
  - Recorded peak performance metrics across models: **Precision 99.94%**, **Recall 99.92%**, **F1-score 99.91%**, and **ROC-AUC 99.34%**, demonstrating robust fault discrimination under severe class imbalance
  - Demonstrated that Bagging-based ensemble methods consistently outperformed Random Forest approaches, with Random-sampler and Balanced Bagging variants excelling across accuracy, recall, and ROC-AUC
  - Proposed a user-driven, real-time fault detection framework enabling early fault identification, reduced AHU downtime, improved energy efficiency, and potential maintenance cost savings

Certifications

- **Machine Learning Specialization**
- **Deep Learning Specialization**

Achievements

- **GATE DA 2025**, AIR 1252 *2025*